METRIC

DOD-D-51486A(EA)
11 July 1986
SUPERSEDING
DOD-D-51486(EA)
10 July 1981

MILITARY SPECIFICATION

DYE MIX, GREEN SMOKE VII (METRIC)

This specification is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a dye mixture composed of Solvent Green 3 and Solvent Yellow 33 dyes.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 6820

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MILITARY

MIL-D-3277 - Dye, Solvent Green 3 (Metric)
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
DOD-D-51485 - Dye, Solvent Yellow 33 (Metric)

STANDARDS

FEDERAL

Fed. Std. No. 313 - Material Safety Data Sheets Preparation and the Submission of

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of nongovernment documents which is current on the date of the solicitation.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

ASTM STANDARDS

D 1895 - Apparent Density, Bulk Factor, and Pourability of Plastic Materials (DOD Adopted)
D 3951 - Commercial Packaging (DOD Adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. Dye Mix, Green Smoke VII shall be a uniformly blended mixture of dyes and shall conform to the composition of table I when tested as specified in 4.2.4.1.

TABLE I. Composition

Material	Percent by weight (pure dye basis)	
	Minimum	Maximum
Dye, Solvent Green 3 conforming to MIL-D-3277	60.75	72.50
Dye, Solvent Yellow 33 conforming to DOD-D-51485	27.60	31.25

3.2 Apparent density. Dye Mix, Green Smoke VII shall have an apparent density of no less than 0.35 gram (g) per milliliter (mL) when tested as specified in 4.2.4.2.

3.3 Material Safety Data Sheets. Material Safety Data Sheets for Dye Mix, Green Smoke VII shall be prepared and submitted by the contractor as specified in Fed. Std. No. 313 (see 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not

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relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the Dye Mix, Green Smoke VII offered for acceptance at one time which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. The minimum lot size shall be 2,270 kilograms. In the event the process is a batch operation (see 6.3), several batches may be mixed together to form a lot provided the batches are subjected to some physical mixing operation intended to make the final product substantially uniform and homogeneous; otherwise, each batch shall constitute a lot. No more than one lot of Solvent Green 3 dye or Solvent Yellow 33 dye shall be used to prepare a lot of Dye Mix, Green Smoke VII.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Three representative specimens of approximately 200 grams (g) each shall be randomly removed from the lot of dye mix offered for acceptance and shall be placed in separate clean, dry containers labeled to identify the lot represented.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled and closed shipping container, ready for shipment. Sample containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Polyethylene liner or closure thereof not as specified (level B only)
(d) Container closure not as specified
(e) Container damaged or leaking
(f) Marking incorrect, missing, or illegible
(g) Desiccant not as specified or missing (level B only)
(h) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

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4.2.4 Tests. Reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Composition.

(a) Standard solution of mixed dyes. Weigh to the nearest 0.1 milligram (mg) exactly 100.0 mg of standard Solvent Yellow 33 (SY-33) and 100.0 mg of standard Solvent Green 3 (SG-3) (see 6.4). Separately dissolve the dyes in methylene chloride (dichloromethane) in 100-mL volumetric flasks; dilute to the volume marks with methylene chloride and mix thoroughly (these stock solutions each contain 1 mg of dye per milliliter). Transfer a 3-mL aliquot of the SY-33 stock solution and a 7-mL aliquot of the SG-3 stock solution to a 12-mL culture tube. Cap the tube and thoroughly mix the solutions (this standard solution contains 0.3 mg of SY-33 and 0.7 mg of SG-3 per milliliter).

(b) Procedure. Weigh to the nearest 0.1 mg exactly 250.0 mg of specimen into a 250-mL volumetric flask. Dissolve in methylene chloride; dilute to the volume mark and mix thoroughly (this solution contains 1 mg of mixed dyes per mL). Using a suitable high pressure-liquid chromatograph, inject 5 microliters each of the solutions of standards and specimen and using a suitable integrator, measure the areas under the SY-33 and SG-3 peaks (retention times are 1.6 and 3.6 minutes, respectively) with the detector set at 290 nanometers. The mobile phase is methylene chloride/acetonitrile (95/5) flowing at 2 mL per minute (1200 pounds per square inch) through a 3.9 by 25 millimeter (mm) guard column filled with 37 to 50 micrometer silica and a 4.6 by 250 mm analytical column containing 5 micrometer silica. Calculate the composition or percents by weight SY-33 and SG-3 as follows:

$$\text{Percent SY-33} = \frac{ABC}{DE}$$

where: A = Percent by weight purity of standard SY-33,
B = Area of specimen peak,
C = Concentration of standard in milligrams per milliliter (0.3),
D = Area of standard peak, and
E = Concentration of specimen in milligrams per milliliter (1.0).

$$\text{Percent SG-3} = \frac{FGH}{JK}$$

where: F = Percent by weight purity of standard SG-3,
G = Area of specimen peak,
H = Concentration of standard in milligrams per milliliter (0.7),
J = Area of standard peak, and
K = Concentration of specimen in milligrams per milliliter (1.0).

Report the percent by weight for each dye as the average of four sets of data from repeated injections.

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4.2.4.2 Apparent density. Determine the apparent density of the specimen in accordance with ASTM D 1895, method A except that the specimen shall be dried to constant weight in an oven at 70° to 75°C prior to testing.

5. PACKAGING

5.1 Unit packing. Dye Mix, Green Smoke VII shall be unit packed level B or commercial as specified (see 6.2).

5.1.1 Level B. Uniform quantities of no less than 65 and no more than 130 kilograms ± 1/4 percent of the marked net weight of dye mix shall be packed in a fiber drum conforming to rule 51, section 2 of the Uniform Freight Classification for a weight limit of over 225 but not over 300 pounds, with an aluminum barrier of 7.62 micrometers minimum thickness incorporated into one of the inner plies of the sidewall. Each drum shall be provided with a close-fitting bag liner formed from polyethylene of 101.6 micrometers minimum thickness conforming to type I, class 1, grade and finish optional of L-P-378. Seams shall be completely heat sealed and shall meet the heat-seal strength requirements of L-P-378. The filled bag shall be closed by heat sealing, tying, or knotting. A minimum of eight units of desiccant conforming to MIL-D-3464 shall be placed on top of the closed liner. Drum closure shall be fully gasketed and shall form a tight seal between the rim and the top of the drum.

5.1.2 Commercial. Uniform quantities of no less than 65 and no more than 130 kilograms ± 1/4 percent of the marked net weight of dye mix shall be unit packed in accordance with ASTM D 3951.

5.2 Packing. Dye mix, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.3 Unitization. Shipping containers shall be unitized in accordance with the applicable requirements of MIL-STD-147.

5.4 Marking. Marking for level B shipments shall be in accordance with MIL-STD-129. Marking for commercial shipments shall be in accordance with ASTM D 3951. All shipments shall be marked to show lot number and date of manufacture. Each container shall be durably and legibly marked with contrasting letters and background to show the net weight of contents. In addition, each container of material shall be marked in accordance with the requirements of public law and regulations. The marking shall include as applicable: name of product; quantity; warning symbol; signal word designating degree of hazard; affirmation statement of hazards; precautionary measures covering actions to be followed or avoided; instructions in case of contact or exposure and notes to physicians; instructions in case of fire, spill, or leak; instructions for handling and storage; and disposal instructions.

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6. NOTES

6.1 Intended use. Dye Mix, Green Smoke VII is intended for use in the manufacture of colored signaling smoke mixtures.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process.

6.4 Dyes of known purity. Standard dyes of known purity for use in 4.2.4.1 may be obtained by authority of the Procuring Contracting Officer (PCO). Alternately, the contractor may prepare his own standard dyes and obtain verification of their purities by authority of the PCO from Commander, U.S. Army Armament, Munitions and Chemical Command, Product Assurance Directorate, Aberdeen Proving Ground, MD 21010-5423.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with Fed. Std. No. 313. The pertinent mailing addresses for submission of data are listed in appendix B of Fed. Std. No. 313.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6820-A089

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-D-51486A(EA)	DYE MIX, GREEN SMOKE VII (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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